

MECHANICAL ENGINEERING PORTFOLIO

Orr Zussman

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B.S. Mechanical Engineering — Expected 2028

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Terrain Emulation Treadmill

Columbia Creative Machines Lab
Research Project Lead

AIM

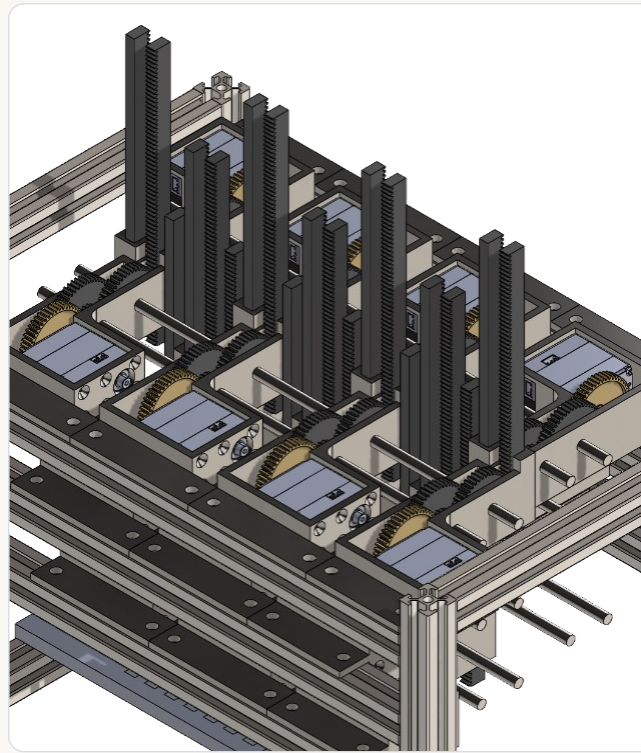
Building a terrain emulating treadmill, with applications in robotic testing, medical rehabilitation, and recreation.

MY CONTRIBUTION

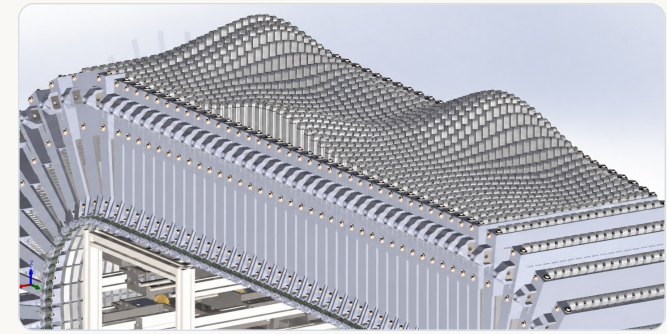
- Own the design process and direct a 4-person team under Prof. Hod Lipson
- Cut servo count 97% (720 to 20) by redesigning the terrain to be “printed” rather than fully actuated
- Designing a next-generation per-segment leadscrew architecture for real-time terrain updates

SKILLS

SolidWorks, FDM 3D printing, waterjet cutting, laser cutting, bandsaw, gear ratios, servo control, Bambu Lab printer maintenance, project management



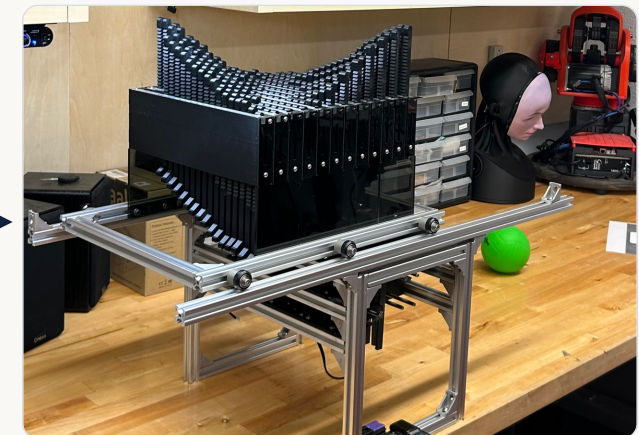
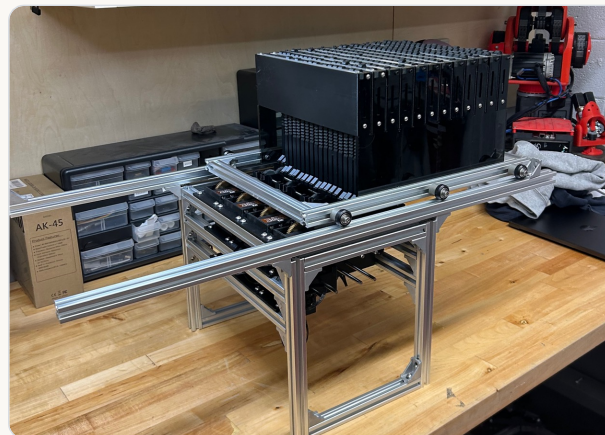
Terrain printing mechanism



Full size treadmill CAD



Ratchet design evolution



Testbed successful emulation

FireDome - Launcher Fabrication

FireDome · Engineering Intern, Interim Job Site Manager
Tel Aviv (Hybrid)

AIM

Assemble FireDome's 8-ton launcher, the platform that deploys its fire-suppression payloads.

MY CONTRIBUTION

- Assembled the launcher with a 4-person crew in 4 weeks
- Made field modifications to resolve post-production design conflicts
- Redesigned the high-voltage capacitor banks at no added cost, cutting assembly man-hours by 50%
- As Interim Job Site Manager, directed heavy equipment (boom crane, manlifts, loaders) and oversaw site safety

SKILLS

SolidWorks, SMAW welding, magnetic drill press, angle grinder, basic hydraulics, basic concrete anchoring, high-voltage work, working at height, directing heavy machinery, safety protocols



Working at height



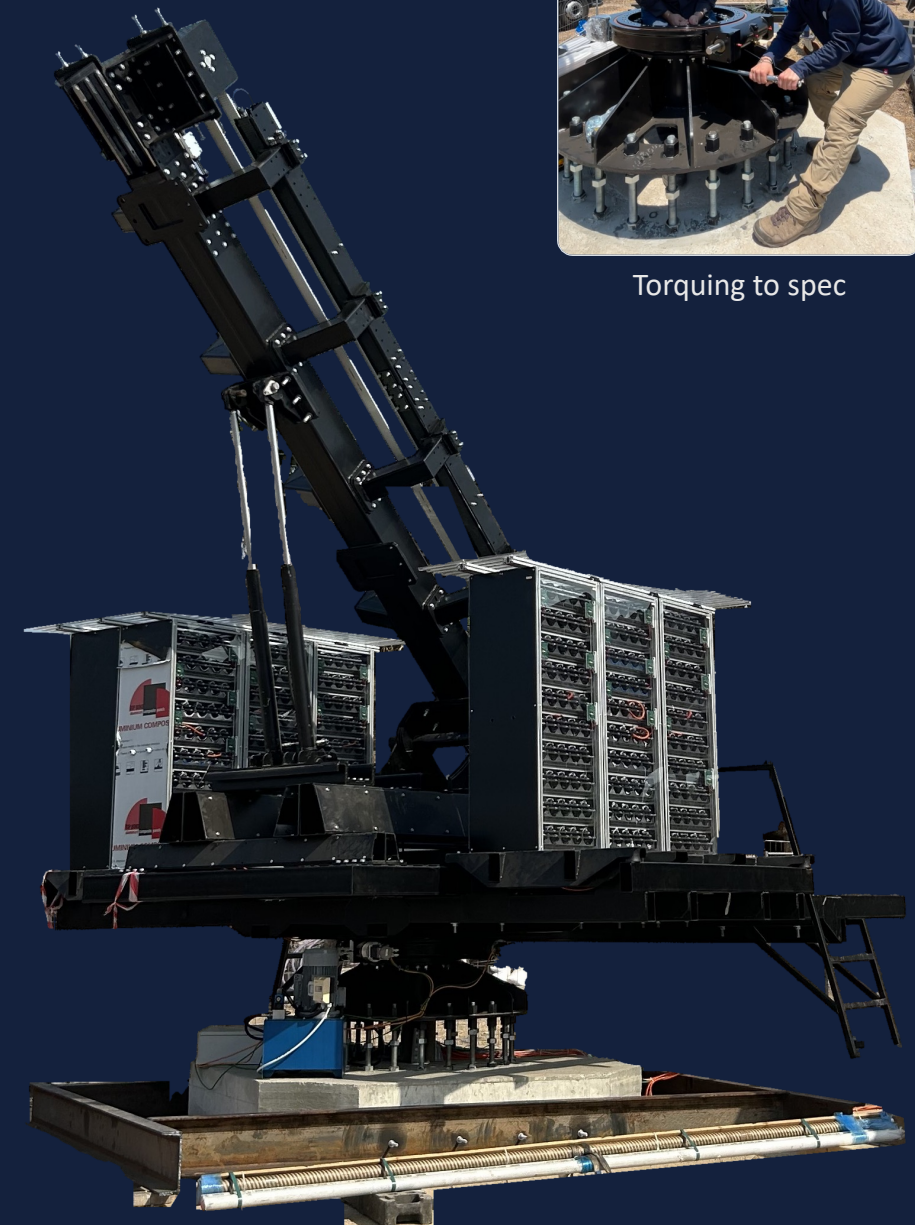
Fabricating capacitor banks



Welding the base



Torquing to spec



ROAR-E — 12lb Combat Robot

Columbia Combat Robotics
Engineering Lead

AIM

Build and compete ROAR-E, a 12 lb combat robot.

MY CONTRIBUTION

- Waterjet cut steel, aluminum and titanium, and formed sheet metal
- 3D printed parts in engineering materials (PA6-GF, TPU)
- Ran pit repairs and turnarounds, restoring the robot to competition readiness under event time limits
- Qualified for the NHRL September Pro Tour
- Now designing a 30 lb robot for the February competition

SKILLS

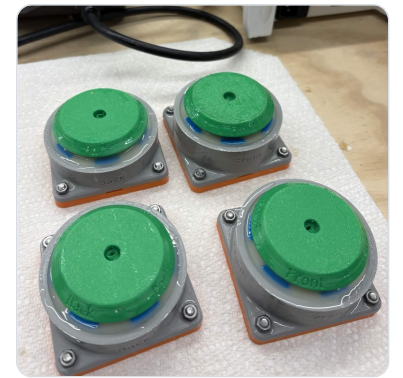
Design for manufacturing, design for repairability, waterjet cutting, basic sheet metal design and forming, composite 3D printing, silicone casting



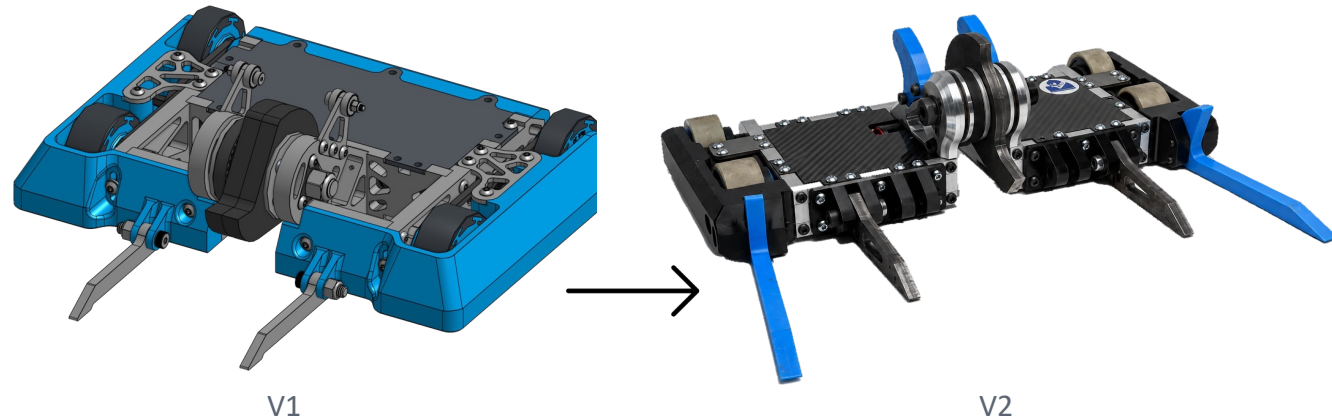
Waterjet-cutting titanium



Aluminum frame



Casting wheels



1. Firey loss



2. Field repairs



3. Victory

Valve & Detonation Mechanism

FireDome · Engineering Intern
Tel Aviv (Hybrid)

AIM

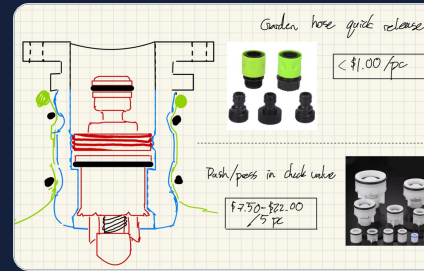
Develop the valve and detonation mechanism for an automatic balloon fill-and-load system.

MY CONTRIBUTION

- Designed a novel valve and quick connect mechanism for a 30 L latex balloon
- Designed timed balloon detonation and airburst mechanism
- Tested and iterated hardware until reliable enough to show investors

SKILLS

Fusion 360, SLS 3D printing, FDM 3D printing, soldering, test planning



1. Initial sketches



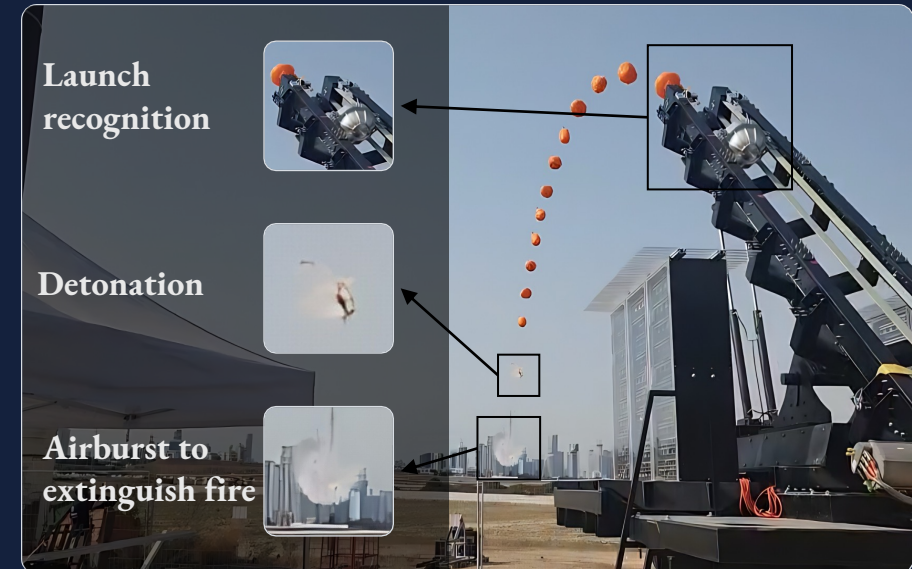
2. Prototyping



3. Testing on 30L balloon

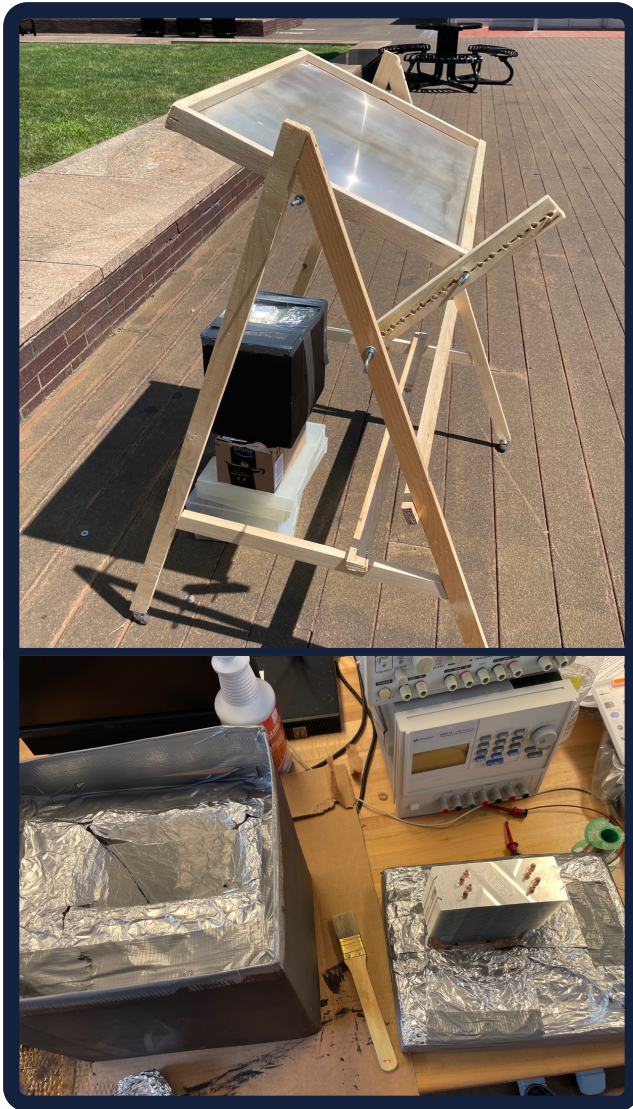


4. Arming launcher

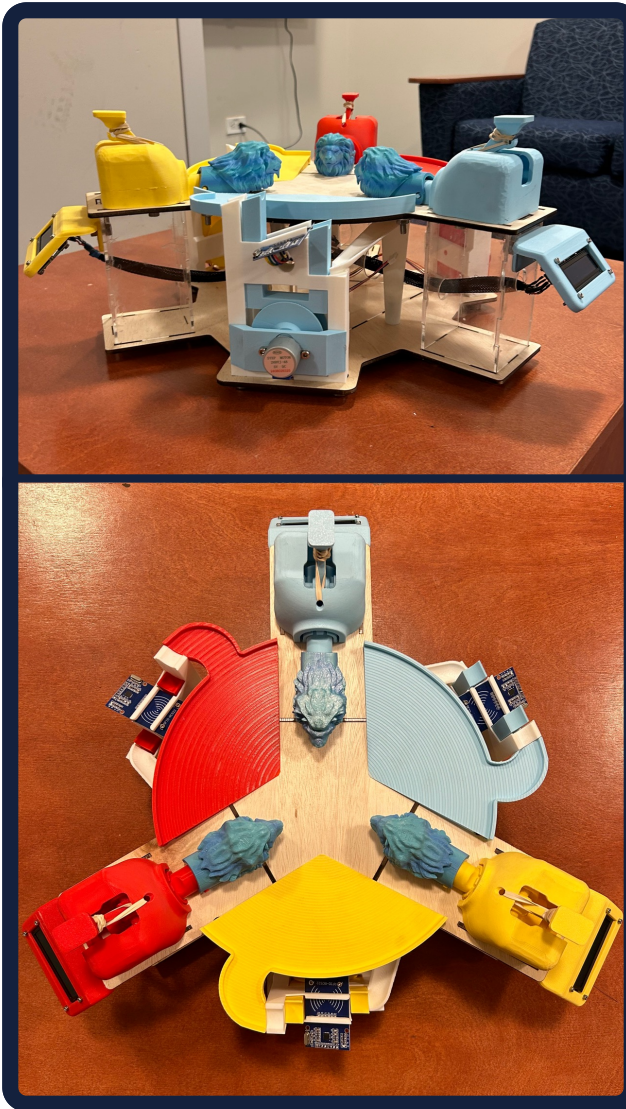


5. Successful test fire

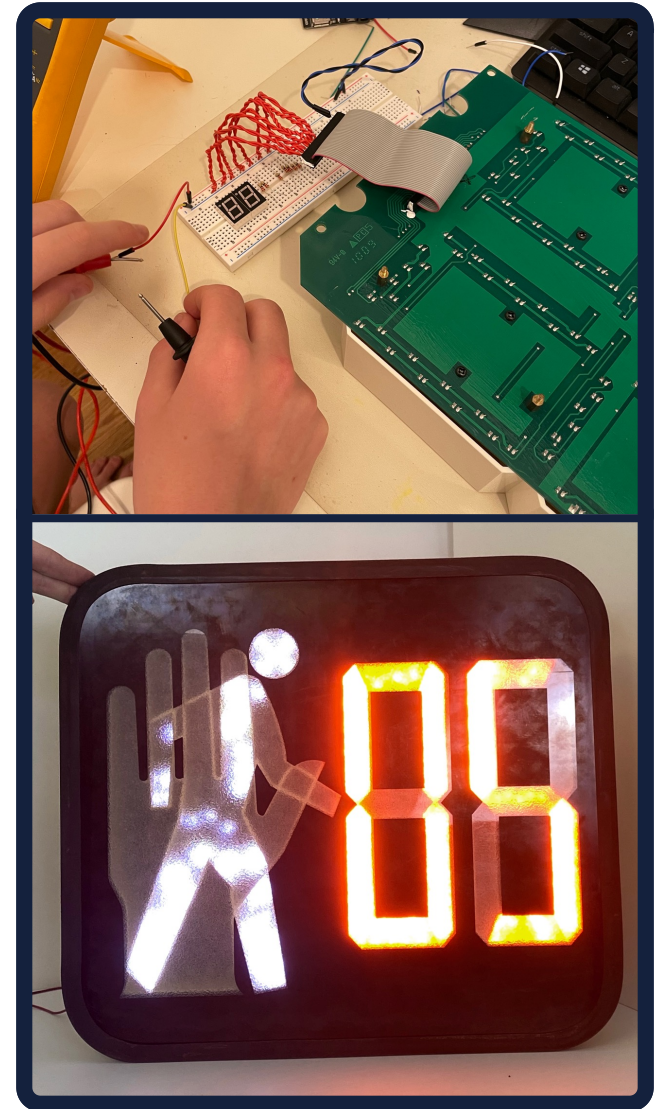
More Projects



350F solar oven



Automated "Hungry Hungry Hippos"



Upcycled traffic light calendar